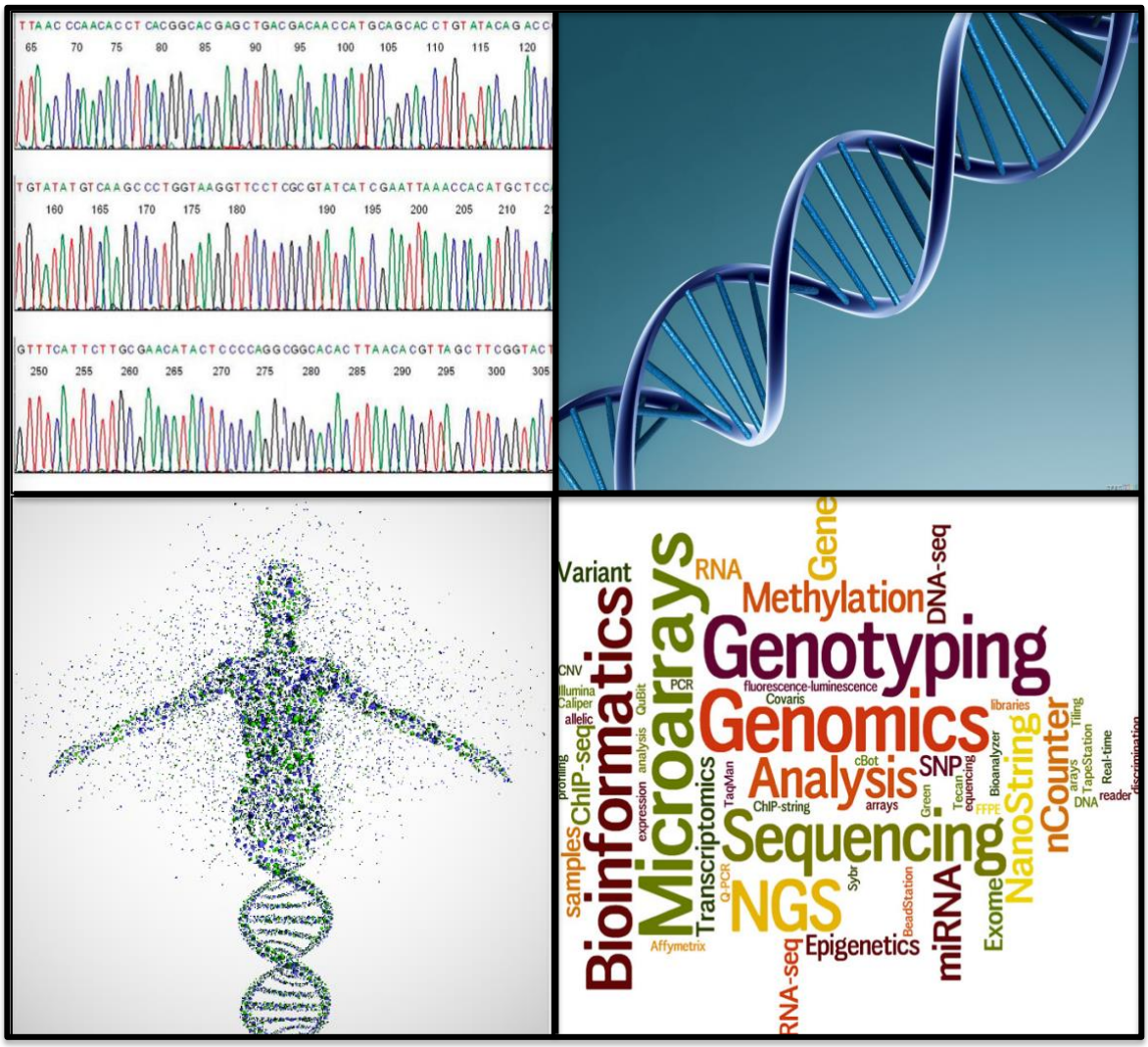
Faculty of Science
School of Life Sciences

UTS

91178 APPLICATIONS OF MOLECULAR BIOLOGY



BIOINFORMATICS MANUAL

Spring 2022

Subject Coordinator: Dr Najah Nassif

SUBJECT MANUAL – PART 1

BIOINFORMATICS COMPUTER EXERCISES (WEEKS 1, 2, 4, 6, 8)

Bioinformatics classes in Spring 2022

This semester, the bioinformatics computer sessions will alternate with the hands-on practical experimental sessions.

The bioinformatics computer classes will run in weeks 1, 2, 4, 6 and 8. These will be followed by the bioinformatics practical test in week 10.

Students in the Monday 9-12 am (01-P1) and 1-4 pm (02-P1) classes will attend their bioinformatics computer classes via Zoom with demonstrators (Zoom link: https://zoom.uts.edu.au/j/81998556447).

Students in the Monday 5-8 pm class will attend their bioinformatics computer classes on campus in CB08.04.004.

BIOINFORMATICS WEEK 1

Work through the first bioinformatics computer exercise on your own. Your demonstrators will go through this with you in the first class in week 2, along with the exercises for week 2.

Before beginning the bioinformatics exercises, the following introductory information will assist you in understanding bioinformatics, and also provides you with the web addresses of some of the essential databases and bioinformatics tools and resources that will be used in the class exercises.

The following introductory information provided below, is also provided in Canvas, within the Bioinformatics module.

Please read through the introductory material before commencing the series of bioinformatics exercises.

INTRODUCTION TO BIOINFORMATICS

Please note: The following material is also provided in the Bioinformatics module in Canvas.

Bioinformatics is the branch of biology that deals with the storage, analysis and interpretation of experimental data. In today's climate of large-scale biology, the datasets can be very large and computers are essential. The instruments of bioinformatics are computers, databases and the statistical tools and algorithms that are used for data analysis. The purpose of bioinformatics is to extract information and identify relationships between datasets. The datasets may be nucleotide or protein sequences, protein structures, gene expression profiles and biochemical or metabolic pathways.

The first nucleotide sequence database to be established was the GenBank database, and, as of 2004, the number of sequences contained in this database was 30 million. This number continues to grow to this day, with over a billion sequences now contained in this database.

With such large numbers of sequences, there is a necessity for the use of computers and computer-based algorithms to analyse them. Overall, bioinformatics is used to assist with sequence retrieval and analysis, molecular modelling, prediction of protein structure and phylogenetic analysis. It forms an essential component of genomics, transcriptomics and proteomics.

Bioinformatics Databases for Storage and Cross-referencing Large Sets of Biological Data

There are many different database types, depending on the nature of the information being stored (e.g. DNA sequences, protein sequences, molecular structures, gene expression data and gel images) and on the manner of data storage (e.g. flat files, tables, graphical representations).

In 2005, the number of different databases available was 719. Each year, the journal 'Nucleic Acids Research' publishes a database issue in which new databases and lists of existing databases of biological data are presented. These can be found at http://nar.oxfordjournals.org/.

A Nucleotide Sequence Databases

Perhaps the most important databases in molecular biology are the ***3 primary sequence databases***. These are:

1. **GenBank:** The GenBank Database is maintained by the US National Centre for Biotechnology Information (NCBI).
2. **DDBJ:** The DNA Databank of Japan.
3. **EMBL:** The EMBL nucleotide sequence databank is maintained by the European Molecular Biology Laboratory (EMBL).

New sequence data can be deposited with any of the 3 databases since they automatically share the data on a daily basis. Also, new data is collected into the databases by automatically scanning the scientific literature. All sequence entries into these databases are extensively annotated (i.e. contain additional descriptive information).

The sequence files retrieved (referred to as flat files) contain not only the sequence but also information on the literature references to the sequence, sequence features and their locations (e.g. the number and positions of exons and introns) and the predicted protein sequence encoded if known.

B Protein Sequence Databases

The main protein sequence databases are the **SWISS-PROT** and the **TrEMBL** databases.

The **SWISS-PROT** database is a collection of confirmed protein sequences that is extensively annotated with information about structure and biological function along with literature references. This is a curated database with high quality data.

The **UniProt database** is the universal protein resource, and, as stated on its website, the database is intended to be both comprehensive and of high quality.

The **TrEMBL** database contains protein sequences derived from the translation of all coding sequences in the EMBL nucleotide sequence database that are not in SWISS-PROT already. However, unlike SWISS-PROT entries, those in TrEMBL are awaiting manual annotation.

The Protein Databank: While SWISS-PROT is the major database of protein sequences, the 3D structures are stored in the Protein Databank (PDB). This is the single world-wide archive of protein structural data derived by X-ray chrystallography, NMR and other techniques.

Further information and online database web locations follow.

Features and Online Locations of the Major Sequence Databases

There are many sequence databases but the four major sequence databases are listed below with their respective online web addresses. One of the most important characteristics of these databases is that they are maintained and kept up to date on a regular basis.

1. The GenBank database at NCBI

Hosted at http://www.ncbi.nlm.nih.gov/

The National Institute of Health (NIH) has achieved a dominant position in collecting biological data of almost any kind. In addition to storing sequence data, NCBI stores almost all kinds of biological sequence related data. PubMed is probably the most used service that NCBI offers to theirs users, together with BLAST, an option for searching for homologous sequences in the entire database.

2. The EMBL database

The EMBL Nucleotide Sequence Database is hosted by the European Bioinformatics Institute (EBI) at the European Molecular Biology Laboratory (EMBL).

Web address: http://www.ebi.ac.uk/embl/

DNA and RNA sequences are directly submitted to the EMBL nucleotide sequence database by individual researchers as well as by genome sequencing projects and patent applications, and the database is produced and maintained in collaboration with both GenBank and the DNA Data Bank of Japan (DDBJ). As mentioned above, the international collection of sequence data is exchanged between EMBL, GenBank and DDBJ on a daily basis and a knowledge of global sequence information can be retrieved from any of the three entries.

3. DNA Data Bank of Japan (DDBJ)

DDBJ (DNA Data Bank of Japan) is a nucleotide sequence database hosted in Japan and accepts DNA submissions mainly from Japanese researchers. They work in close collaboration with GenBank and EMBL and the three databases store almost identical data. DDBJ also provides various search and analysis tools through their website.

Web address: http://www.ddbj.nig.ac.jp/

4. The UniProt protein sequence database

The UniProt database is the universal protein resource, and, as stated on its website, the database is intended to be both comprehensive and of high quality: "The mission of UniProt is to provide the scientific community with a comprehensive, high-quality, and freely accessible resource of protein sequence and functional information."

Web address: http://www.uniprot.org/

At the UniProt website, data has been divided into three classifications: (1) core data, (2) supporting data and (3) information and sequences may be searched in these three categories.

The UniProt Knowledgebase, UniProtKB, contains translations of the coding sequences submitted to EMBL, GenBank and DDBJ as well as all publicly available protein sequences.

WEEK 1 - BIOINFORMATICS 1

USING THE NCBI WEBSITE
DATABASES AND DATABASE MINING PART 1

Please note: Work through this exercise on your own (during week 1) and your demonstrators will go through it with you in the first bioinformatics class in week 2.

Introduction to the Computer Laboratories

Overall, the computer exercises undertaken over the next 5 weeks are designed to allow you to learn and understand the use of computers for the mining of sequence and other databases to obtain sequence and other biological information from publicly available databases (some of which have been introduced above).

Through the retrieval of biological data in the form of DNA and protein sequence information (and corresponding annotations), students will learn to read and interpret biological data and identify aspects of gene structure for real sequence examples. These computer sessions extend student knowledge and understanding beyond the formal lecture material and develop the ability to search and evaluate scientific literature.

Background to the Computing Exercises

The bioinformatics exercises scheduled for each week may be carried out on any DNA or protein sequences of interest to illustrate the principles being taught in the exercises. The sequences used are chosen for illustrative purposes.

For the purposes of the computer exercises, the sequences of choice will vary from week to week and may include genes important to cell function, genes implicated in various human diseases, genes encoding enzymes involved in cell metabolism and genes involved in regulating the cell cycle and cancer development. Mutations in genes that are important for cell function and survival can lead to cellular phenotypic changes that are characteristic of several medical conditions and disorders including cancer, diabetes, cystic fibrosis, muscular dystrophy and many others.

Many genes are known to exist in many different species of organisms, where they essentially carry out similar functions. Comparing the sequences of the same gene in different organisms can tell us how closely related the 2 organisms are by the level of sequence similarity observed (e.g. human and gorilla gene sequences show up to 99% similarity for some gene sequences).

The computer labs will give you the opportunity to investigate genes, gene sequences and protein sequences using computer-based bioinformatics tools. Students will gain proficiency in using the bioinformatics tools to solve problems requiring the analysis of sequence data.

Competence in the use of the bioinformatics tools will be assessed in the bioinformatics practical test scheduled for week 8.

WEEK 1 - BIOINFORMATICS 1

EXERCISE 1

Using the NCBI Website to Obtain Literature and Information on Genes of Interest

Aims

Initially, the structure of the GenBank database and the bioinformatics resources at NCBI will be investigated and then utilised to obtain sequence information and other knowledge of some human genes of interest.

Use of the Entrez Browser and the NCBI Databases

(A) Use of the GenBank database and NCBI is one of the most popular methods for investigating and obtaining information related to genes, proteins and relevant sequence data which have been characterised to date.

- Using a web browser (e.g. Firefox, Google Chrome or Safari. Avoid using Internet Explorer), navigate to the NCBI home page at the following address: http://www.ncbi.nlm.nih.gov/ (also provided above).

- Click on the "About the NCBI" link on the home page.

  This will bring up an information page containing material about the resources available on NCBI under 4 different headings or sections.

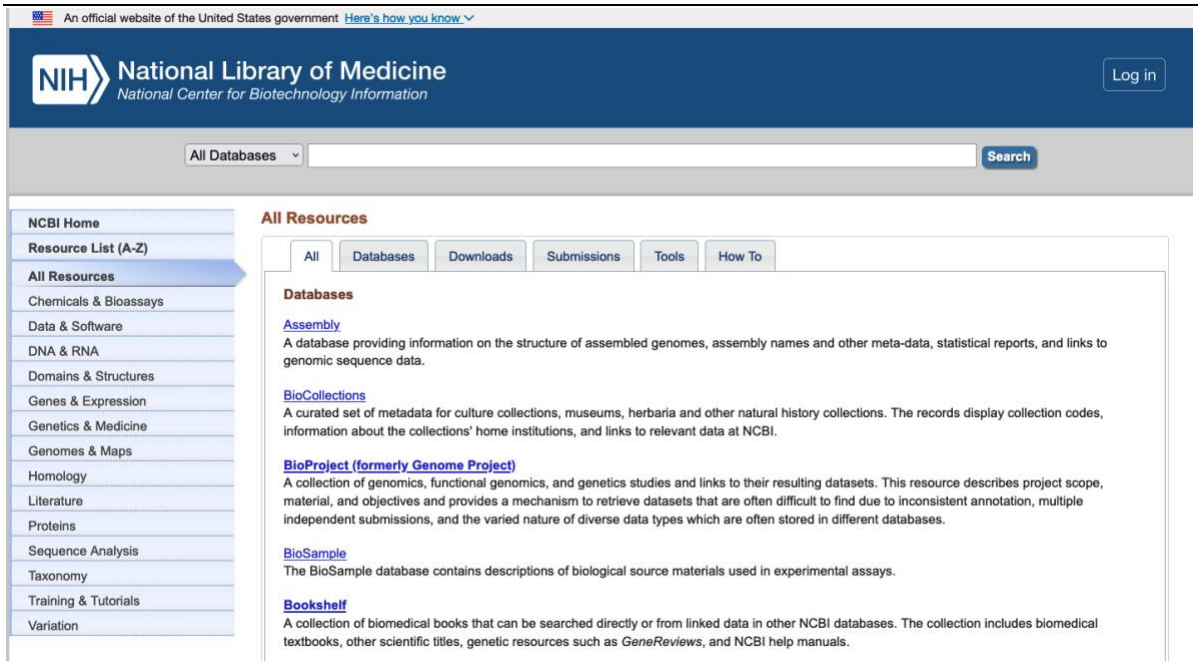
  The section entitled 'Databases and Tools' provides more detailed information about the NCBI and its services.

  Read through this section at their own pace to learn about the different bioinformatics resources and tools available at NCBI. Working through this section will help you through the exercises that will follow.

  You will be able to navigate back to this page for help at any time if needed.

  NOTE: To ensure students are on the correct web link, a snapshot of the relevant page is shown below.

  The 'Tools' tab contains a list, with descriptions, of the tools and functions available at NCBI for bioinformatics analyses.



NOTE: There are also a number of video tutorials with instructions on 'how to' use some of the different resources contained at NCBI. These can be found by selecting the 'how to' tab (see above figure) or by navigating directly to the following address:

http://www.ncbi.nlm.nih.gov/education/tutorials/

HINT: You can always refer back to the tutorials section of NCBI if you forget how to do a particular analysis or need some information.

(B) The Entrez database will now be used to identify resources and obtain information relating to some genes of interest.

1. Navigate to the NCBI homepage again and select '**PubMed'.** This should take you the following web link: http://www.ncbi.nlm.nih.gov/pubmed.

   PTEN is a tumour suppressor gene known to be mutated in many different cancer types including brain, breast, endometrial, prostate and ovarian cancers among many others.

2. From the search dropdown menu, select 'All Databases' and search for information using the keyword '**PTEN'** (i.e. type this word into the search bar).

   This will identify a large number of literature references and other resources on this topic that are maintained by NCBI.

3. To look at all literature references relating to PTEN, click on the '**PubMed'** link.

   The next screen will show the documents and reference materials found in the search, along with details of the references.

   Individual references can be selected by clicking on their name. This will bring up more detailed information about the selected reference.

In most cases, a published reference of interest may be downloaded directly from PubMed if freely available.

Limiting a search

The scope of a search may be limited in many ways such as by including other terms and phrases (e.g. including Boolean operators such as "PTEN **and** review") or limiting the dates of the search or selecting specific authors. This will reduce the large number of references detected in a search.

This is advisable when you know exactly what you are searching for.

BIOINFORMATICS 1

ACTIVITY 1

1. Navigate to the NCBI homepage again and select '**PubMed'.**

2. From the search dropdown menu, select 'All Databases' and search for information using the keyword '**PTEN'** (i.e. type this word into the search bar).

3. Use the information obtained from your search to complete the following results Table 1. The first 2 entries have been completed for you *(i.e. at the time of the preparation of this subject manual and could now be changed as more data is deposited daily)*.

Results Table 1

Information Resource Available at NCBI	No. of Records for PTEN
Books	614
PubMed	21,790
PubMed Central	
ClinVar (i.e. clinical variants)	
OMIM	
Nucleotide	
SNP	
Conserved Domains	
Protein	
Structure	
Genome	

4. Repeat the above search using the following search terms in turn:

   (a) 'Human PTEN' (note, we could have used the species name '*homo sapiens'* in place of human)

   (b) 'Mouse PTEN' (note, we could have used the species name *mus musculus* in place of mouse)

   (c) '*C. elegans* PTEN' (*C. elegans* is a nematode worm)

   (d) 'Gorilla PTEN' (note, we could have used the species name '*gorilla gorilla'* in place of gorilla)

5. Use the information obtained from each of the above searches to complete the following results table 2. The first entry has been completed for you.

   *[**Please note:** the pre-filled numbers in Table 2 are correct at the time of the preparation of this subject manual and could now be changed as more data is deposited daily.]*

Results Table 2

Information Resource Available at NCBI	**PTEN**	**Human PTEN**	**Mouse PTEN**	**Gorilla PTEN**	***C. elegans* PTEN**
Books	**506**	**283**	**188**	**0**	**45**
PubMed					
PubMed Central					
ClinVar (i.e. clinical variants)					
OMIM					
Nucleotide					
SNP					
Conserved Domains					
Protein					
Structure					

Question 1 **From the information in results Table 2, which of the 4 species has the greatest number of literature references relating to PTEN?**

6. From the search results for 'Human PTEN', click on the OMIM link to select it. A list of diseases/disorders related to PTEN will now show up.

Question 2 **Scanning through the first 10 entries in the list, note down 3 human diseases or disorders in which PTEN is implicated.**
Disease/disorder 1:
Disease/disorder 2:
Disease/disorder 31:

7. Using the Entrez browser (as above), search 'All Databases' using the search term 'insulin'.

8. Use the results of the search to fill in the appropriate details into results Table 3 below. You will also need to transfer your results for PTEN (from above) for comparison.

   *[**Please note:** the pre-filled numbers in the Table 3 are correct at the time of the preparation of this subject manual and could now be changed as more data is deposited daily.]*

Results Table 3

Information Resource from NCBI	No. of Records for PTEN	No. of Records for Insulin
Books	**506**	**12,847**
PubMed		
PubMed Central		
ClinVar (i.e. clinical variants)		
OMIM		
Nucleotide		
SNP		
Conserved Domains		
Protein		
Structure		
Genome		

The insulin gene and protein sequences were determined and characterised long before the characterisation of the gene and protein sequences of PTEN (approximately 1960's and 1997 respectively).

Question 3

(A) In light of the above facts, do you expect there will be a greater number of information resources for insulin?

(B) Does the information in results table 3 support your expectation?

(C) If yes, how did you come to this conclusion?

Question 4 **(A) Which of the 2 sequences (insulin or PTEN) shows the greater level of sequence variation?** *[Hint: look at the SNP results for the 2 genes. SNPs are single nucleotide polymorphisms which are single base changes in the sequence that occur between individuals.]*
(B) Are you able to quantify the level of sequence difference (e.g. is the sequence variation of one gene twice that of the other, etc.)?

9. Repeat the search for **'PTEN'** using 'All Databases'.

10. When you have the results available again, select the **'Gene'** link. This will take you to a list of PTEN gene sequences from various organisms.

11. Select the first entry in the list (*homo sapiens* PTEN) by clicking on it.

12. If you now scroll through the information to the section entitled **'Genomic context'**, you will see information about the chromosomal location of human PTEN.

    The notation '10q23.3' provided means the PTEN gene is on chromosome 10 (at position 23.3 of the long arm of chromosome 10).

Question 5

Using the same search protocol, can you find the chromosomal location of the following genes in Table 4? Include your answers in results Table 4 below.

[Hint: Read the list carefully and make sure you select the correct gene before noting the chromosomal location.]

Results Table 4

Gene Name	**Chromosomal Location**
Homo sapiens beta actin	
Homo sapiens Glyceraldehyde-3-phosphate dehydrogenase (GAPDH)	
Homo sapiens Glucose-6-phoshate dehydrogenase (G6PDH)	
Homo sapiens Beta globin	
Homo sapiens p53	
Homo sapiens Cyclin dependent kinase 4 (CDK4)	
Homo sapiens Retinoblastoma 1 tumour suppressor (RB1)	
Homo sapiens Collagen type 1	
Homo sapiens Breast cancer 1 (BRCA1)	
Homo sapiens Breast cancer 2 (BRCA2)	
Homo sapiens Creatine kinase M-type	
Homo sapiens Growth hormone 1	

Now that you know how to find gene-related information (e.g. chromosomal location, involvement in disease, etc.) using the NCBI database, you can do this for any gene that you may have an interest in or that you may see appearing in news and current affairs programs. This often occurs when new genes involved in disease (e.g. muscular dystrophy, cancer, hypertension, diabetes, mental illness, cystic fibrosis and other chronic diseases) are discovered and reported to the public.

In the next exercise, you will move a step further and learn how to extract and analyse the nucleotide sequence of a gene of interest.

BIOINFORMATICS 1

DATABASES AND DATA MINING

Introduction

As outlined in the introductory material above, there are a number of databases for the storage of the large amounts of biological data generated by many researchers worldwide. Thus information about individual genes, sequences and their products are stored in these databases. These databases are easily accessible through the web at particular web locations (refer to introduction above). Individual gene sequence data and information can be obtained through database searches.

The most important contribution that the sequence databases make to the scientific community is making the sequences themselves accessible.

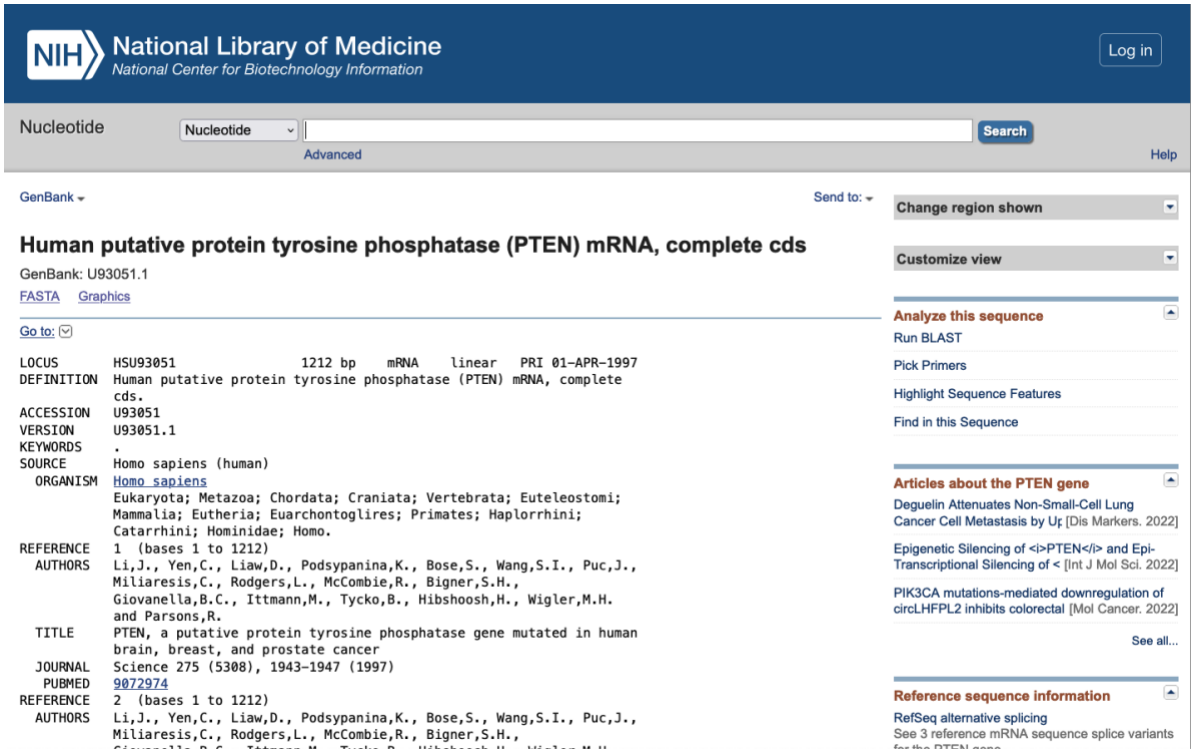
When querying a database, the information output obtained is typically referred to as a 'flat file' which is a text file containing gene or protein sequence information with annotations. The format of flat file outputs will differ between databases, but the essential information they contain is identical, regardless of database source.

Aim

The purpose of this exercise is to help students develop an understanding of the contents and information contained within a typical flat file obtained from the GenBank database through the NCBI website.

BIOINFORMATICS 1
ACTIVITY 2 – PART 1

1. Navigate to the NCBI homepage as before.
2. Select the **'Nucleotide'** database from the drop down menu on the search bar at NCBI.
3. Search for the entry '**HSU93051'** by typing the word into the search bar. This should identify a single entry (U93051) in the Nucleotide sequence database corresponding to the PTEN sequence. A snapshot of the result is shown below.



Clicking on any of the underlined web links within the flat file, will display additional information about the selected sequence.

This flat file contains many fields, which are listed on the left side of the page from the top down.

For example, the locus is the Name given to the entry and the accession number is a database-specific identifier for that sequence which will always find this entry, and so on.

If the PubMed entry number is selected, the browser will navigate to the reference link for the gene of interest at the PubMed site.

For **U93051 and the information stored in the corresponding flat file, answer the following questions.**

Question 1 **What species is the PTEN sequence derived from?**

Question 2 **(A) What is the GenBank accession number for the PTEN coding sequence?**
(B) What is meant by the term accession number in this context?

Question 3

In which journal(s), and in what year(s), was the paper describing the discovery of this sequence published?

[Hint: Read carefully, there may be more than one paper describing the discovery of a sequence. Sometimes two research laboratories may make the same discovery almost simultaneously.]

Journal Name(s)	**Year(s)**

Question 4 **What is the length of the entire mRNA sequence deposited in the GenBank database for PTEN (U93051)?**

Question 5

(A) What is the length of the coding sequence (CDS) in base pairs?

(B) On the basis of the length of the coding sequence, calculate the number of codons present in the CDS.

[Hint: Remember the genetic code and how many bases make up a codon.]

Question 6

(A) From question 5, predict how many amino acids should be present in the gene product.

(B) Looking at the flat file, how many amino acids are in the protein sequence?

Question 7

At what positions in the sequence are the start and stop codons of the CDS located?

Start codon position:

Stop codon position:

Question 8

Are any parts of the 5' and 3' untranslated regions of the mRNA represented in the sequence?

Question 9

Based on the information provided in the flat file, is there any evidence for the presence of introns in this gene?

Please note:

You will be able to check your answers for this exercise in next week's (i.e. week 2) classes. Your demonstrators will work through this exercise with you before commencing the exercises set for week 2.

This activity will be continued and expanded in the next bioinformatics session in week 2.

WEEK 2 - BIOINFORMATICS 2

DATABASES AND DATABASE MINING PART 2
SEQUENCE SIMILARITY SEARCHING (BLAST)

BIOINFORMATICS 2
ACTIVITY 2 – PART 2

1. Select the 'Nucleotide' tab as before and search for the entry '**AF067844'** by typing the letters and numbers into the search bar. This should identify a single entry (AF067844) in the Nucleotide sequence database corresponding to the PTEN genomic sequence.

   The gene or genomic sequence of PTEN is the sequence as it exists in the genome before transcription.

   On first glance it is obvious that the length of the gene is much longer than that of the mRNA represented in the previous flat file (U93051).

   For **AF067844 and the information stored in the corresponding flat file, answer the following questions.**

Question 1 **What species is the PTEN sequence derived from?**

Question 2 **What is the GenBank accession number for the PTEN genomic sequence and from which chromosome location is it derived?**
Accession No:
Chromosome location:

Question 3

In which journal(s), and in what year(s), was the paper describing the discovery of this gene sequence published?

[Hint: Read carefully, there may be more than one paper describing the discovery of a sequence. Sometimes two research laboratories may make the same discovery almost simultaneously.]

Journal Name(s)	Year(s)

Question 4 **Looking at the flat file, how many amino acids are in the protein sequence?**

Question 5 **Are any parts of the 5' and 3' untranslated regions of the mRNA represented in the sequence?**
If so, at what positions do they occur?
5' UTR Position:
3' UTR Position:

Question 6

Based on the information provided in the flat file, is there any evidence for the presence of introns in this gene?

Question 7

If your answer to question 6 was yes, how many exons are present in the gene?

Question 8

Use the description in the flat file to fill in the exon number and position in the table below.

Exon number	Gene position

Question 9

How would you account for the difference in length between the PTEN gene sequence and the PTEN mRNA sequence?

2. While viewing the flat file, selecting 'Highlight sequence features' on the right hand side menu will highlight any sequence features as you scroll through the flat file.

   In this case, the coding regions are highlighted.

   Try it yourself. Select 'Highlight sequence features' and scroll through the flat file. You will notice certain sequences are now highlighted and details are provided to indicate the reason for the highlighting.

3. Go back to the flat file for **U93051**. The protein sequence of PTEN has been submitted to the protein database and clicking on the 'Protein id' hyperlink will take you to the protein flat file entry for PTEN.

4. From the protein database entry (AAC51183), answer the following questions.

Question 10

What is the length (in amino acids) of the PTEN protein?

BIOINFORMATICS 2
ACTIVITY 2 – PART 3

For the following flat file for the human myoglobin gene, highlight the features that have been annotated (as indicated in the flat file).

Your highlights should, at least, include the following:

- coding sequence
- start codon
- stop codon

```
LOCUS       BC014547                1191 bp    mRNA    linear   PRI 15-JUL-2006
DEFINITION  Homo sapiens myoglobin, mRNA (cDNA clone MGC:13548 IMAGE:4244838),
             complete cds.
ACCESSION   BC014547
VERSION     BC014547.1
KEYWORDS    MGC.
SOURCE      Homo sapiens (human)
ORGANISM    Homo sapiens
            Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi;
            Mammalia; Eutheria; Euarchontoglires; Primates; Haplorrhini;
             Catarrhini; Hominidae; Homo.
REFERENCE   1  (bases 1 to 1191)
   AUTHORS   Strausberg,R.L., Feingold,E.A., Grouse,L.H., Derge,J.G.,
             Klausner,R.D., Collins,F.S., Wagner,L., Shenmen,C.M., Schuler,G.D.,
             Altschul,S.F., Zeeberg,B., Buetow,K.H., Schaefer,C.F., Bhat,N.K.,
             Hopkins,R.F., Jordan,H., Moore,T., Max,S.I., Wang,J., Hsieh,F.,
             Diatchenko,L., Marusina,K., Farmer,A.A., Rubin,G.M., Hong,L.,
             Stapleton,M., Soares,M.B., Bonaldo,M.F., Casavant,T.L.,
             Scheetz,T.E., Brownstein,M.J., Usdin,T.B., Toshiyuki,S.,
             Carninci,P., Prange,C., Raha,S.S., Loquellano,N.A., Peters,G.J.,
             Abramson,R.D., Mullahy,S.J., Bosak,S.A., McEwan,P.J.,
             McKernan,K.J., Malek,J.A., Gunaratne,P.H., Richards,S.,
             Worley,K.C., Hale,S., Garcia,A.M., Gay,L.J., Hulyk,S.W.,
             Villalon,D.K., Muzny,D.M., Sodergren,E.J., Lu,X., Gibbs,R.A.,
             Fahey,J., Helton,E., Ketteman,M., Madan,A., Rodrigues,S.,
             Sanchez,A., Whiting,M., Madan,A., Young,A.C., Shevchenko,Y.,
             Bouffard,G.G., Blakesley,R.W., Touchman,J.W., Green,E.D.,
             Dickson,M.C., Rodriguez,A.C., Grimwood,J., Schmutz,J., Myers,R.M.,
             Butterfield,Y.S., Krzywinski,M.I., Skalska,U., Smailus,D.E.,
             Schnerch,A., Schein,J.E., Jones,S.J. and Marra,M.A.
CONSRTM     Mammalian Gene Collection Program Team
  TITLE     Generation and initial analysis of more than 15,000 full-length
            human and mouse cDNA sequences
   JOURNAL  Proc. Natl. Acad. Sci. U.S.A. 99 (26), 16899-16903 (2002)
    PUBMED  12477932
REFERENCE   2  (bases 1 to 1191)
   CONSRTM  NIH MGC Project
   TITLE    Direct Submission
   JOURNAL  Submitted (24-SEP-2001) National Institutes of Health, Mammalian
            Gene Collection (MGC), Bethesda, MD 20892-2590, USA
   REMARK   NIH-MGC Project URL: http://mgc.nci.nih.gov
 COMMENT    Contact: MGC help desk
            Email: cgapbs-r@mail.nih.gov
            Tissue Procurement: CLONTECH
            cDNA Library Preparation: CLONTECH Laboratories, Inc.
            cDNA Library Arrayed by: The I.M.A.G.E. Consortium (LLNL)
            DNA Sequencing by: Institute for Systems Biology
            http://www.systemsbiology.org
            contact: amadan@systemsbiology.org
            Anup Madan, Jessica Fahey, Erin Helton, Mark Ketteman, Anuradha
```

```
            Madan, Stephanie Rodrigues, Amy Sanchez and Michelle Whiting
            Clone distribution: MGC clone distribution information can be found
            through the I.M.A.G.E. Consortium/LLNL at: http://image.llnl.gov
            Series: IRAL Plate: 19 Row: g Column: 18
            This clone was selected for full length sequencing because it
            passed the following selection criteria: matched mRNA gi: 44955876.

            Differences found between this sequence and the human reference
            genome (build 36) are described in misc_difference features below
            and these differences were also compared to chimpanzee genome
            (build 2).
FEATURES
            Location/Qualifiers
      Source          1..1191
                      /organism="Homo sapiens"
                      /mol_type="mRNA"
                      /db_xref="taxon:9606"
                      /clone="MGC:13548 IMAGE:4244838"
                      /tissue_type="Skeletal Muscle"
                      /clone_lib="NIH_MGC_81"
                      /lab_host="DH10B"
                      /note="Vector: pDNR-LIB"
      gene            1..1191
                      /gene="MB"
                      /gene_synonym="MGC13548"
                      /gene_synonym="PVALB"
                      /db_xref="GeneID:4151"
                      /db_xref="HGNC:HGNC:6915"
                      /db_xref="MIM:160000"       misc_difference 1..4
                      /gene="MB"
                      /gene_synonym="MGC13548"
                      /gene_synonym="PVALB"
                      /note="4 bases at the 5' end do not align to the human
                      genome."
      CDS             77..541
                      /gene="MB"
                      /gene_synonym="MGC13548"
                      /gene_synonym="PVALB"
                      /codon_start=1
                      /product="myoglobin"
                      /protein_id="AAH14547.1"
                      /db_xref="GeneID:4151"
                      /db_xref="HGNC:HGNC:6915"
                      /db_xref="MIM:160000"
                      /translation="MGLSDGEWQLVLNVWGKVEADIPGHGQEVLIRLFKGHPETLEKF
                      DKFKHLKSEDEMKASEDLKKHGATVLTALGGILKKKGHHEAEIKPLAQSHATKHKIPV
                      KYLEFISECIIQVLQSKHPGDFGADAQGAMNKALELFRKDMASNYKELGFQG"
      misc_difference 250
                      /gene="MB"
                      /gene_synonym="MGC13548"
                      /gene_synonym="PVALB"
                      /note="'A' in cDNA is 'G' in the human genome; no amino
                      acid change. The chimpanzee genome agrees with the human
                      genomic sequence and not the cDNA."
      misc_difference 280
                      /gene="MB"
                      /gene_synonym="MGC13548"
                      /gene_synonym="PVALB"
                      /note="'T' in cDNA is 'C' in the human genome; no amino
                      acid change. The chimpanzee genome agrees with the human
                      genomic sequence and not the cDNA."
      misc_difference 1075..1191
                      /gene="MB"
                      /gene_synonym="MGC13548"
                      /gene_synonym="PVALB"
                      /note="polyA tail: 117 bases do not align to the human
                      genome."
ORIGIN
```

```
   1 ggggaaaccc cagctgttgg ggccaggaca cccagtgagc ccatacttgc tctttttgtc
  61 ttcttcagac tgcgccatgg ggctcagcga cggggaatgg cagttggtgc tgaacgtctg
 121 ggggaaggtg gaggctgaca tcccaggcca tgggcaggaa gtcctcatca ggctctttaa
 181 gggtcaccca gagactctgg agaagtttga caagttcaag cacctgaagt cagaggacga
 241 gatgaaggca tctgaggact taaagaagca tggtgccact gtgctcaccg ccctgggtgg
 301 catccttaag aagaaggggc atcatgaggc agagattaag cccctggcac agtcgcatgc
 361 caccaagcac aagatccccg tgaagtacct ggagttcatc tcggaatgca tcatccaggt
 421 tctgcagagc aagcatcccg gggactttgg tgctgatgcc caggggggca tgaacaaggc
 481 cctggagctg ttccggaagg acatggcctc caactacaag gagctgggct tccagggcta
 541 ggcccctgcc gctcccaccc ccacccatct gggccccggg ttcaagagag agcggggtct
 601 gatctcgtgt agccatatag agtttgcttc tgagtgtctg ctttgtttag tagaggtggg
 661 caggaggagc tgaggggctg gggctggggt gttgaagttg gctttgcatg cccagcgatg
 721 cgcctccctg tgggatgtca tcaccctggg aaccgggagt ggcccttggc tcactgtgtt
 781 ctgcatggtt tggatctgaa ttaattgtcc tttcttctaa atcccaaccg aacttcttcc
 841 aacctccaaa ctggctgtaa ccccaaatcc aagccattaa ctacacctga cagtagcaat
 901 tgtctgatta atcactggcc ccttgaagac agcagaatgt ccctttgcaa tgaggaggag
 961 atctgggctg ggcgggccag ctggggaagc atttgactat ctggaacttg tgtgtgcctc
1021 ctcaggtatg gcagtgactc acctggtttt aataaaacaa cctgcaacat ctcaaaaaaa
1081 aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa
1141 aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa a
```

NOTES

BIOINFORMATICS 2: SEQUENCE SIMILARITY SEARCHING

ACTIVITY 3

Introduction

In bioinformatics, **BLAST,** for **B**asic **L**ocal **A**lignment **S**earch **T**ool is an algorithm for comparing primary biological sequence information, such as the amino acid sequences of different proteins or the nucleotides of DNA sequences. A BLAST search enables a researcher to compare a query sequence with a library or database of sequences and identify library sequences that resemble their own query sequence (above a certain statistical threshold).

A **sequence alignment** is a way of arranging the sequences of DNA, RNA or protein to identify regions of similarity that may be a consequence of either functional, structural or evolutionary relationships between the sequences.

In a scenario where scientists may have sequenced a new gene and are unsure of a function, use of a BLAST search of the new sequence against a database of sequences can provide a great deal of information. Based on the similarity of the gene sequence with previously characterised gene sequences, clues as to a possible gene identity and gene function may emerge.

Before Class preparation

Before starting, navigate to http://www.ncbi.nlm.nih.gov/guide/all/#tools_ and familiarise yourself with the various tools for searching databases.

Work through the on-line tutorial on Blast searching. The tutorials are within the education section accessible from the home page (http://www.ncbi.nlm.nih.gov/education/tutorials/). There are both web-based and video tutorials available.

BIOINFORMATICS 2

ACTIVITY 3 – PART 1

1. Having worked through the tutorial, describe the differences between the BlastN, BlastP, BlastX, TBlastN and TBlastX search programs. Write your answers in the spaces provided below.

BLASTN

BLASTP

BLASTX

TBLASTN

TBLASTX

BLASTN Analysis of PTEN

2. Search the Nucleotide database of NCBI for the GenBank entry for homo sapiens PTEN (U93051).

3. Once you have located the flat file, select 'Run BLAST' from the right-hand side menu to compare the mRNA sequence of PTEN against all gene sequences contained in the GenBank database.

4. In the blast window that comes up, click on the 'BLAST' button towards the bottom of the page to start the search.

5. From the BLASTN output, answer the following questions.

Question 1

Name five other species that show sequence similarity with the human PTEN sequence and show the % identity determined by the BLAST search.

No.	Species Name	% Identity
1		
2		
3		
4		
5		

Question 2 **For a gene with functions important to cell survival, did you find the results from question 1 surprising? If so, why? If not, why?**

6. The above BLAST search was carried out after a database search using the flat file for the sequence. In the case where you have an unknown sequence, you will not be able to use this approach.

   In this case, you will need to type in, or copy and paste, the sequence of interest into the search box of the BLASTN search engine.

7. For the next exercise, navigate to the BLASTN server at http://blast.ncbi.nlm.nih.gov/Blast.cgi.
   Select 'nucleotide blast' from the menu.

Now copy and paste (or simply type in) the following human sequence into the BLASTN search box.

```
ATGATCGCCG CGCAGCTCCT GGCCTATTAC TTCACGGAGC TGAAGGATGA
CCAGGTCAAA AAG
```

Now search the database to pick up sequences similar to the sequence of interest.

[Note: Selecting any of the matching sequences in the list will take you to a view of the actual matches between the 2 sequences at the nucleotide level.]

From the BLASTN results, answer the following questions.

Question 3

Based on the BLAST results, what is the most likely identity of the unknown gene sequence?

[Hint: It is of human origin.]

Question 4

What is the usual function of the protein encoded by this gene?

Question 5

Name 3 other species that show sequence similarity with the unknown (or query) sequence and show the % identity determined by the BLAST search.

No.	Species Name	% Identity
1		
2		
3		

8. For the final part of this exercise, run a blastn search on the following human DNA sequence.

```
agggaacccc ttacctggaa tctggaatca gcctcttctc tgatgaccct
gaatctgatc cttctgaaga cagagcccca gagtcagctc gtgttggcaa
cataccatct tcaacctctg cattgaaagt tccccaattg aaagttgcag
aatctgccca gagtccagct gctgctcata ctactgatac tgctgggtat
aatgcaatgg aagaaagtgt gagcagggag aagccagaat tgacagcttc
aacagaaagg gtcaacaaaa gaatgtccat ggtggtgtct ggcctgaccc
cagaagaatt t
```

Question 6

(A) Based on the BLAST results, what database sequence is your unknown sequence most similar to?

(B) Is it possible that this may be the identity of your unknown sequence (i.e. the input sequence is a part of the total sequence)?

(C) From these results, could Blast be used to identify a gene from an unknown sequence?

Question 7 **Are there any sequence homologs in any other species in the top 100 list?**
If so, which species?
If not, why do you think this may be the case?

9. Click on the accession number of the matching sequence at the top of the list.

   This will take you to the flat file corresponding to the gene of interest.

   Once in the flat file, select the 'Gene' link from the right hand side menu.

   From the summary information provided, answer the following question.

Question 8 **What is the putative function of the following query gene sequence?**

BIOINFORMATICS 2

ACTIVITY 3 – PART 2

The BLAST search tool is most commonly used for searching a query sequence against the sequence databases to identify sequences that are similar.

However, in some cases, you may have 2 sequences (either DNA, RNA or protein) that you would like to align with each other to determine their 'similarity' to each other.

In this case, you will need to enter each of the sequences (or their accession numbers if known) into the individual entry areas in the BLAST search tool.

Aligning sequences to determine similarity

1. Navigate to the BLAST search tool page at: https://blast.ncbi.nlm.nih.gov/Blast.cgi?PAGE_TYPE=BlastSearch

2. Select the **'align two or more sequences'** box.

   You will see that an additional sequence box is now visible.

3. Enter the accession number **BC005255** (corresponding the human insulin mRNA/cDNA sequence) into the top box on the BLAST search page.

4. Enter the accession number **NM_001109772** (corresponding to the porcine, or pig, insulin mRNA/cDNA sequence) into the low box on the BLAST search page.

5. Click on the BLAST button to begin the search and wait for the output.

Question 1 **From the BLAST output, what is the % identity between the 2 sequences?**

From the alignment output, you can see how similar the two sequences are.

Question 2

Which part(s) of the two sequences show the greatest similarity?

(Hint: Look at the flat files for each sequence and see which areas correspond to coding or non-coding regions.)

6. Repeat the sequence similarity search using the following two accession numbers.

   BC005255 (human insulin mRNA / cDNA sequence)

   AH002844 (human insulin gene sequence)

   You can search for and find the flat files for these 2 sequences while waiting for the BLAST output to be completed.

Question 4

From the BLAST output, how many regions of high sequence similarity are observed?

Question 5

From the sequence information in the flat files, what type of sequence do these sequence regions represent?

Question 6

What type of sequences might be present in the sequence regions that are not showing any similarity (i.e. not represented in the output)?

WEEK 4 - BIOINFORMATICS 3

RNA SPLICING AND ALTERNATIVE TRANSCRIPTS

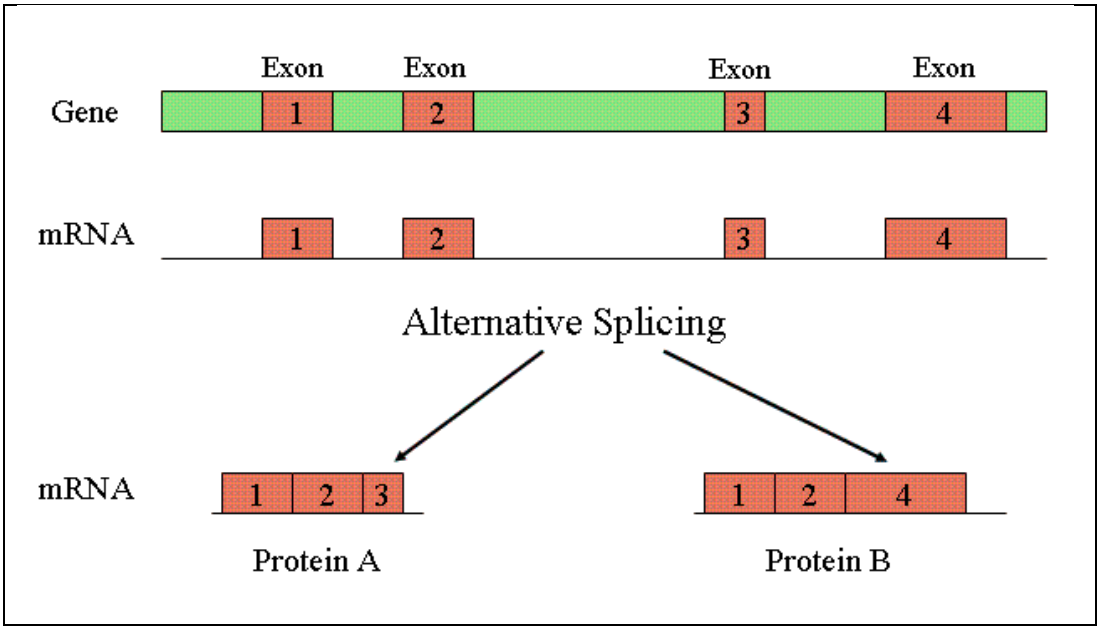
EXPASY AND PROGRAMS FOR ANALYSING PROTEIN STRUCTURE AND FUNCTION

RNA SPLICING AND ALTERNATIVE TRASCRIPTS

Introduction: mRNA Maturation

Genes are transcribed to generate an RNA copy (primary transcript) that is then modified post transcriptionally to produce the mature messenger RNA (mRNA) that will then be used for translation to produce the protein product. In eukaryotes, the modification of the primary RNA transcript typically involves 3 steps: (i) the addition of a 5' cap, (ii) the removal of any intron sequences and the re-joining (splicing) of exons and (iii) the addition of a poly(A) tail, to generate the mature mRNA sequence.

Alternative Splicing: The mRNA generated from some genes can be edited in a number of different ways, resulting in the generation of splice variants. That is, different polypeptides are synthesised from a single gene (i.e. by altering the order in which exons are spliced together). The example shown below illustrates the generation of 2 polypeptides from the alternative splicing of a single mRNA transcript.



Aims

This exercise aims to develop and extend student knowledge of mRNA splicing (step ii of the RNA maturation process) and alternate splicing using specific bioinformatics resources and the human PTEN mRNA sequence (U93051) as the example input sequence.

BIOINFORMATICS 3
ACTIVITY 4

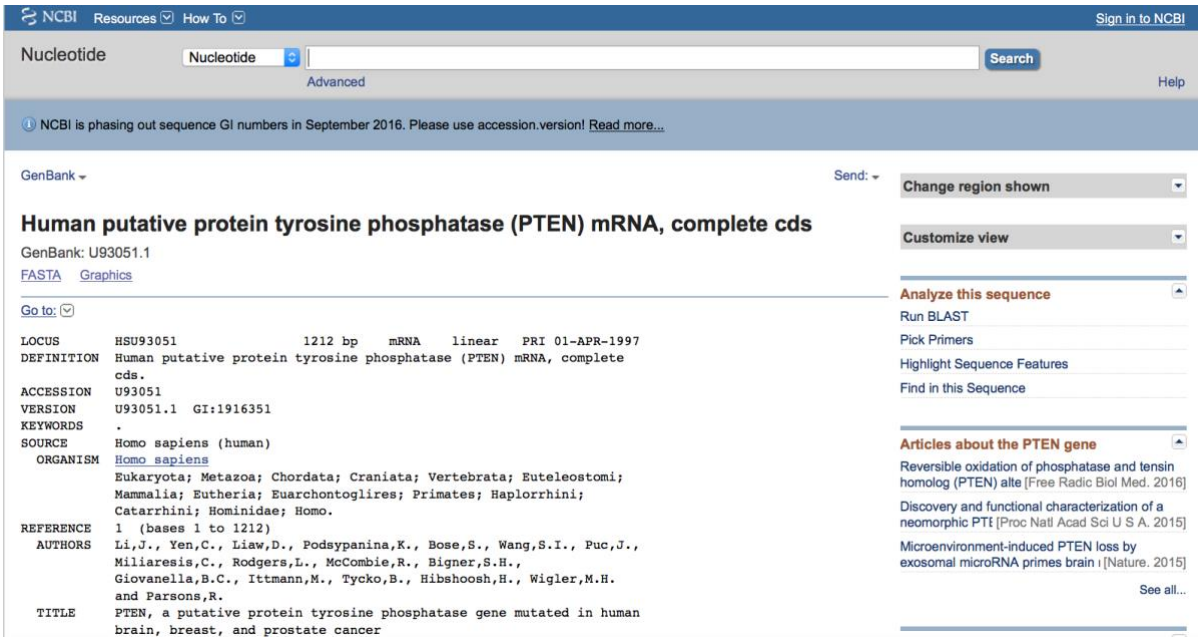
There are a number online databases and resources allowing the visualisation and prediction of RNA splicing events for an input RNA sequence. The following web resources are available for such tasks and will be useful for this activity.

The PTEN mRNA sequence (U93051) will be used as the input sequence to some of these programs, for generating genome maps of the RNA sequence and to visualise the positions of exons and introns.

- The **Ensembl Genome Browser:** http://www.ensembl.org/index.html
  This allows mapping of mRNA, protein and predicted gene sequences onto the human, mouse, rat and other genomic sequences with a view of the intron and exon structure of the sequence.

  Ensembl is a joint scientific project between the European Bioinformatics Institute (EBI) and the Wellcome Trust Sanger Institute, which was launched in 1999 in response to the imminent completion of the Human Genome Project. The aim of Ensembl is to provide a centralised resource for geneticists, molecular biologists and other researchers studying the genomes of our own species and other vertebrates. Ensembl is one of several well known genome browsers for the retrieval of genomic information.

  Sequences in the GenBank database may be linked directly to the Ensembl Genome Browser from the NCBI web site by selecting the **'Graphics'** link at the top of the sequence entry (see the following snapshot from NCBI below).



- **UCSC Bioinformatics Database:** http://www.genome.ucsc.edu/
  Select **'Genome Browser'** from the menu set on the left hand side of the screen. Inputting a sequence will allow mapping of the mRNA sequence onto the human, mouse and rat genomes.

- **NCBI AceView:** http://www.ncbi.nlm.nih.gov/IEB/Research/Acembly/index.html
  Allows mapping of mRNA sequences on to the Human and *C. elegans* genomic sequences.

1. Navigate to the NCBI website and search for PTEN mRNA sequence by its accession number U93051.

2. Once in the flat file, select the 'Gene' link for the menu on the right.

3. Now select 'Ensembl' from the menu on the right (under 'Links to other resources). You will now be taken to the Ensembl browser and you will see the exons mapped onto the genomic sequence of PTEN.

Question 1 **From the Ensembl output, how many introns and exons are predicted to be present in the human PTEN gene?**
No. of exons:
No. of introns:

4. To view the different alternative transcripts that may occur, select the 'Show Transcript Table' and a list will appear.

Question 2 **How many alternatively spliced mRNAs are shown in the table?**

5. Now navigate to the NCBI AceView and again search the human database entry for PTEN.

6. From the output obtained determine the following:

Question 3 **How many introns and exons are predicted to be present in the human PTEN gene and how many alternatively spliced mRNAs are actually known to exist?**
No. of exons:
No. of introns:
No. of alternatively spliced transcripts:

9. Compare your predictions with data stored in GenBank (i.e. the relevant flat files) and find the accession numbers for any alternatively spliced mRNAs of human PTEN.

Write your notes here.

BIOINFORMATICS 3

EXPASY AND PROGRAMS FOR ANALYSIS OF PROTEIN STRUCTURE AND FUNCTION

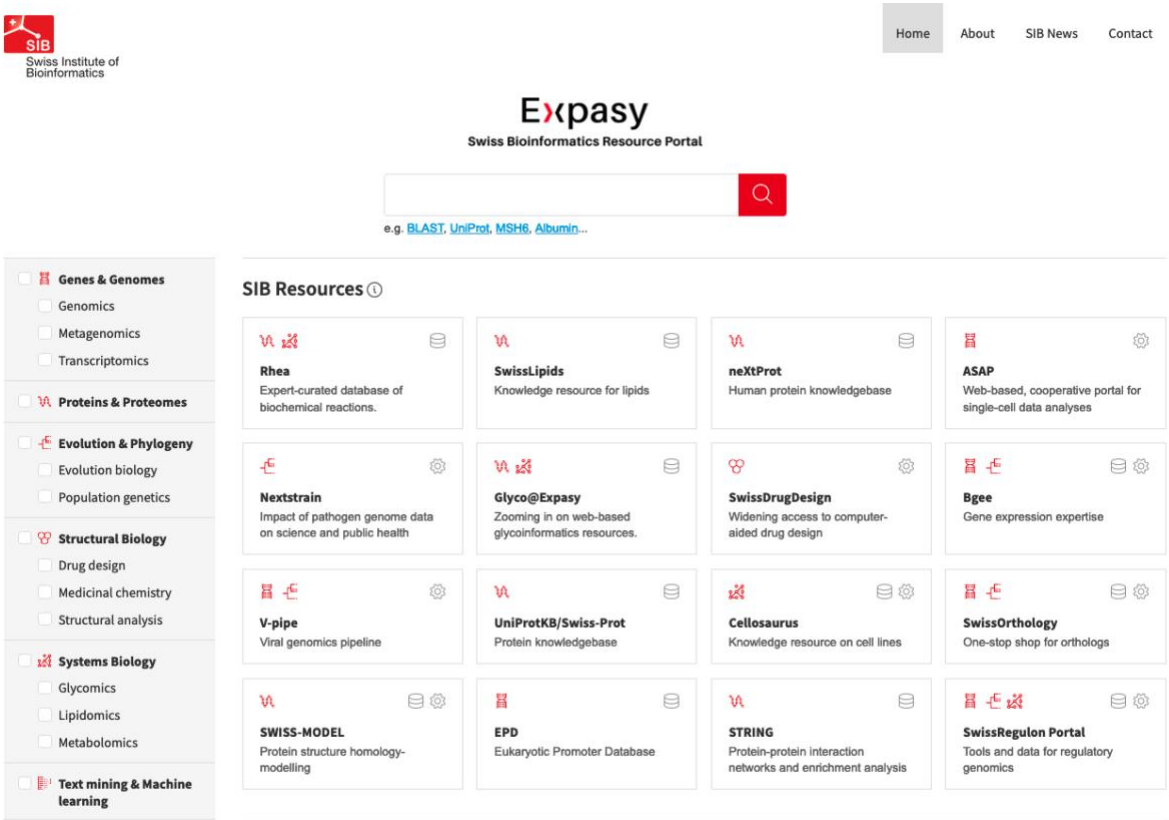
Introduction

This exercise, will make use of the protein sequence databases to retrieve a protein sequence of interest and then use the resources at the ExPASy portal to carry out an analysis of the sequence to locate the position of a specific sequence motifs of interest in the protein.

ExPASy (Expert Protein Analysis System) is a bioinformatics resource portal operated by the Swiss Institute of Bioinformatics (SIB). It is an extensive and integrative portal accessing many scientific resources, databases and software tools in different areas of life sciences. Scientists can access a wide range of resources in many different domains, such as proteomics, genomics, phylogeny/evolution, systems biology, transcriptomics etc.

A single web portal provides a common entry point to a wide range of resources and the portal features a search function across selected resources. The portal is aimed for both expert users and for people that are not familiar with a specific domain in life sciences: in particular, the new Web interface provides visual guidance for newcomers to ExPASy (see snapshot of the portal below.

ExPASy web address: https://www.expasy.org/



Aims

The main aim of this exercise is to introduce students to the idea that computer algorithms can be used to determine protein structural features, predict potential phosphorylation and glycosylation sites, molecular weight, hydrophobicity, pI and many other features based on the protein sequence.

Most significantly, computer algorithms are used also to predict the structure and function of proteins based on the knowledge of known protein sequences and their corresponding functional information. It should be stressed that the results of such predictive analyses are hypothetical and provide information which may be further explored and confirmed by experiments that can determine protein structure (e.g. NMR, X-ray chrystallography) and function. Nevertheless, such results can be informative, especially in the absence of other information regarding a particular protein sequence.

There are many databases and servers dedicated to protein sequences and their analysis throughout the world (refer to the introductory material to the bioinformatics section). In this exercise, the ExPASy portal will be utilised. The sequence of interest to be used is the human PTEN protein sequence (accession no. AAD13528).

In part 1 of this exercise, the NetPhos tool will be used predict phosphorylation sites within the PTEN protein and in part B, the and ClustalW tool will be used align the human PTEN protein sequence with PTEN protein sequences from other other organisms to determine similarities and differences among them.

BIOINFORMATICS 3
ACTIVITY 5 – PART 1

1. Navigate to the NCBI website and search the 'Nucleotide' database for U93051 to obtain the flat file relating to the PTEN sequence.

2. From the flat file, you will see a link to the protein id number for the corresponding protein sequence for PTEN.

3. Click on the protein id number to be taken to the protein entry. In this flat file you will find information relating to the human PTEN protein sequence.

4. From the information contained in the flat file, answer the following questions

Question 1 **What are the 2 regions identified in the PTEN protein and at what positions in the protein do they occur?**
Region 1:
Region 2:

5. Go to the **ExPASy** Proteomics Server at http://www.expasy.org/ and select the 'Resources A-Z' tab and investigate the programs available for protein sequence analysis (arranged in alphabetical order).

6. To get started and test the tools available, select the 'NetPhos' tool. This tool will scan the PTEN protein sequence for potential phosphorylation sites. Phosophorylation of proteins occurs on Ser (S), Thr (T) and Tyr (Y) amino acid residues.

   NOTE: Should the NetPhos tool be unavailable at the ExPASy server, navigate to the alternernative NetPhos tool (v3.1) site at the following web address: https://services.healthtech.dtu.dk/service.php?NetPhos-3.1

   The PTEN protein is known to be phosphorylated on certain Ser and Thr residues near the C-terminal end of the protein (the specific sites can be found in the literature).

   The output of this scan will predict which Ser and Thr and amino acid residues in the PTEN protein are possible sites of phosphorylation.

Remember from your biochemistry learning that phosphorylation of proteins can make them either more active or less active. This is the same for PTEN. This protein acts as a phosphatase enzyme.

Hence the finding of potential phosphorylation sites has implications for how the activity and function of this protein might be regulated.]

7. Copy the PTEN protein sequence from the Genpept flat file (accession no. AAC51183) into the query box of the 'NetPhos' tool.

8. Limit the scan to serine and threonine phosphorylation only and begin the scan.

   From the output answer the following questions:

Question 2 **Did the program predict any potential phosphorylation sites?**

Question 3 **How many sites of Serine phosphorylation were there?**

Question 4 **How many sites of Threonine phosphorylation were there?**

Question 5

Where did the sites of Serine and Threonine phosphorylation occur in the PTEN protein? Fill in the positions in the results table below (the first entry has been completed for you).

Phosphorylated residue	**Position in PTEN sequence**	**Phosphorylated residue**	**Position in PTEN sequence**
Ser	**57**		

Phosphorylated residue	Position in PTEN sequence	Phosphorylated residue	Position in PTEN sequence

Question 6

Based on the literature on PTEN phosphorylation at Serine residues, do any of your predicted Ser phosphorylation sites coincide with experimentally determined Ser phosphorylation sites? If so, which Ser residues were they?

[Hint: As a start, search PubMed for papers on 'PTEN AND phoshorylation'.

9. Repeat the phosphorylation scan after selecting Tyr phosphorylation to scan for possible Tyrosine phosphorylation sites.

Question 7

Were there any potential Tyrosine phosphorylation sites predicted for PTEN?

BIOINFORMATICS 3

MULTIPLE SEQUENCE ALIGNMENT (NUCLEOTIDE AND PROTEIN)

Multiple sequence alignment: An Introduction

A **multiple sequence alignment** is a sequence alignment of three or more biological sequences, generally protein, DNA or RNA. In many cases, the input set of query sequences are assumed to have an evolutionary relationship by which they share a lineage and are descended from a common ancestor. From the resulting multiple sequence alignment, sequence homology can be inferred and phylogenetic analysis can be conducted to assess the possible shared evolutionary origins of the sequences.

In bioinformatics **Clustal** is a series of widely used computer programs for carrying out multiple sequence alignments. There have been many versions of the Clustal program including **Clustal** (the original software), **ClustalV** (a rewrite of the original Clustal package), **ClustalW, ClustalX** (this version has a graphical user interface) and **Clustal Omega.**

Aim

Using the **ClustalW** program at **ExPASy**, we will carry out a multiple sequence alignment to visualise the similarities and differences between a set of sequences of interest. The input sequences will be **PTEN protein sequences** derived from **5 different species** (human, gorilla, chimpanzee, mouse and rat).

BIOINFORMATICS 3

ACTIVITY 5 – PART 2

1. To carry out a multiple sequence alignment, the first requirement is to obtain the sequences of interest.

2. Navigate to the NCBI website and search the '**Protein**' database 'human PTEN'. This will bring the human (*homo sapiens*) PTEN protein sequence flat file.

3. Select the 'FASTA' link to obtain the protein sequence in the FASTA format.

   Note: the FASTA format is as follows:

   >Sequence name
   Sequence

4. Select the FASTA format sequence (i.e. name and sequence together) and paste into Word or other text editor program.

5. Now repeat the process for the other sequences in turn.

Gorilla (*gorilla gorilla*) PTEN
Cimpanzee (*pan paniscus*) PTEN
Rat (*rattus norvegicus*) PTEN
Mouse (*mus musculus*) PTEN

6. Once you have obtained all 5 sequences in FAST format, you will need to join the files together end to end into one large file (see below).

   >Name of Sequence 1
   sequence 1
   >Name of Sequence 2
   sequence 2
   >Name of Sequence 3
   sequence 3
   >Name of Sequence 4
   sequence 4
   >Name of Sequence 5
   sequence 5

7. Navigate to the ExPaSy server and select ClustalO from list of programs.

8. Select the text from your merged sequences file and paste the sequences into the input sequence box of the clustalO program.

9. Run ClustalO

IMPORTANT

In the event that ClustalO at ExPaSy is not working, you may use Clustal Omega at http://www.ebi.ac.uk/Tools/msa/

Once you select '**launch Clustal Omega**', you will be greeted with a sequence input box where you will paste your merged sequences file.

Then click 'submit' to begin the alignment.

Your output will appear soon after.

Apart from the alignment, this portal also provides the output of a phylogenetic tree as well.

Notes about the output

The "*"- asterisk denotes that residue at that position are identical.

The ":"- colon indicates that residues at that position have similar properties (e.g. V and L).

The "."- dot indicates residues are more or less similar.

If there is no mark, it denotes that there is no common property.

From your output, answer the following questions.

Question 1

Do all the PTEN protein sequences from the different organisms have the same length?

Question 2

How many amino acid differences are there among the 5 sequences?

Question 3

What do you estimate as the level of similarity between the 5 PTEN protein sequences?

[This can be expressed as a percentage.]

Question 4

What might be some possible reasons for the level of similarity observed for this protein sequence among the different organisms and species?

10. Repeat the Clustal alignment after the addition of a 6th PTEN protein sequence, that of *Drosophila melanogaster* PTEN.

Question 5 **From this new output, has the level of sequence similarity changed or has it remained the same?**
Question 6 **What did this indicate about the Drosophila sequence compared to the other 5 PTEN sequences?**

WEEK 6 - BIOINFORMATICS 4

MOLECULAR EVOLUTION AND PHYLOGENETIC ANALYSIS USING RIBOSOMAL RNA AND OTHER SEQUENCES

Introduction

Some genes encode RNAs that are not translated into protein but instead, carry out their function at the RNA level. Such RNAs are termed non-coding RNAs (as they do not code for a protein) or structural RNAs. One class of well know structural RNAs are the ribosomal RNAs that make up important components of the ribosome, the machinery of translation.

Most eukaryotic cells contain four distinct ribosomal RNA types that are encoded by genes (rDNA) which are repeated many times in the genome. Such tandemly repeated sequences make up their own class of repetitive sequences. Ribosomal RNA and rDNA sequences have been used for many years now to investigate the genetic relationships amongst organisms (phylogeny) through the analysis of sequence similarities within the ribosomal RNA sequences between different species. As such, these sequences represent a very important class of molecules to study. Indeed they are extensively used to construct a tree of life. (look up!)

Molecular Phylogenetic Analysis

Molecular phylogenetic analysis is the use of macromolecular sequences (such ribosomal RNA sequences) to reconstruct the evolutionary relationships between organisms. The extent of difference between homologous DNA, RNA, or protein sequences in different organisms is used as a measure of how much these organisms have diverged from one another evolutionarily.

Steps of a molecular phylogenetic analysis:

1. Decide what organisms/sequence/gene/region to examine.
2. Determine the sequence(s) experimentally.
3. Identify homologous residues.
4. Carry out phylogenetic analysis: compare sequences to determine relationships.

Deciding on a sequence for analysis: Some sequences are better than others - the most important factors to consider in a sequence for phylogenetic analysis are:

1. **Clock-like behavior**: i.e. sequence divergence in the gene between two organisms should be proportional to how long ago they diverged. Clock-like behavior depends mostly on functional constancy of the sequence - function change leads to large, directed sequence change.

2. **Phylogenetic range:** The sequence must be present in all of the organisms to be analysed and must have retained its structure and function in these organisms. The gene

must have enough variation in sequence to give information but must be similar enough so that homologous residues can be recognised and lined up (alignment).

3. **No horizontal transfer:** This means that the gene must be acquired only by inheritance, not by transfer from another organism.

4. **Must have a large existing data set** with which to compare your sequences. In other words, other sequences of that gene should be in GenBank!

18S rRNA

The gene encoding the 18S rRNA in the large subunit of the ribosome is a good choice for phylogenetic analysis because:

* It is present in all cells.
* It has exactly the same function in all cells.
* It is conserved enough in sequence and structure of be readily and accurately aligned.
* Horizontal transfer of rRNA genes is absent or rare.
* There is a large database (>100,000) of aligned sequences available

Aim

The study of ribosomal RNA sequences and their use for phylogenetic analysis is the basis of this exercise. The 18S rRNA sequence (accession number U03069) will be used for phylogenetic analysis in this exercise.

BIOINFORMATICS 4
ACTIVITY 6

1. Using the NCBI database locate the 18S rDNA sequence that has the accession number U03069.

Within the flat file obtained, determine whether the sequence present is an RNA sequence or a DNA sequence. How can you tell?

2. Display the sequence as a FASTA file by selecting the FASTA link at the top of the sequence.

3. In the FASTA file format, note the header at the top of the sequence. Now copy the entire sequence entry, with header, for further use in this exercise.

4. Using the BLASTN option, search the total nucleic acids database with the sequence entry from accession number U03069.

From the BLAST output file, name two other species with sequences similar to that of U03069.
1.
2.

Do the sequences from the two species share sequence similarity? If so, are they homologous? Why?

Determining the evolutionary ancestry of DNA sequences, and hence the species they are derived from, is complex and first involves performing multiple sequence alignments of the sequences of interest.

From multiple sequence alignments, between homologous genes derived from different organisms, genetic relationships between the sequences can then be calculated using different mathematical equations and algorithms. The evolutionary relationships are then inferred from the sequence similarities and differences.

For more information and instructions on conducting phylogenetic analyses, navigate to the following website: http://phylogeny.lirmm.fr/phylo_cgi/index.cgi (for a beginner's guide to conducting a phylogeny study).

Use the 18S rDNA sequence (accession number AY730405) from the NCBI database in the following exercises.

5. Navigate to http://phylogeny.lirmm.fr/phylo_cgi/index.cgi and select the 'Explore your sequence neighbours' option.

6. Select BLASTN and paste the 18S rDNA (AY730405) sequence into the box provided (use FASTA format).

7. Submit the sequence search to the default database at NCBI.

8. There will be a small wait, after which the program will return a phylogenetic tree showing the nearest neighbours (to the sequence of interest) found by the BLASTN search of the sequence database.

   The neighbours are linked together in a tree structure, joined by branches that show the relationships between the sequences.

   This is not a true phylogenetic tree, but illustrates the basic ideas that phylogeny involves the steps of sequence alignment (of the sequences of interest) and then prediction of the relationships of the sequences from the alignments.

From the output tree structure, which species of organisms do you think are closely related to the organism from which the sequence AY730405 is derived?

The phylogeny website also allows you to perform more detailed phylogenetic analyses.

For this part of the exercise, we will determine the genetic relationships between a number of different organisms based on their 18S rRNA sequences.

9. Find the following 18S rDNA sequences in NCBI and cut and paste each one of them into a word file (similar to last week) in a suitable format (i.e. in FASTA format).

   The 18S rDNA sequences to look up are:

   AY730405 *Dientamoeba fragilis*
   AF293056 *Histomonas meleagridis*
   DQ174297.1 *Monocercomonas colubror*um
   DQ855405.1 *Caduceia versatilis*
   AY886869.1 *Tetratrichomonas buttreyi*
   AY245121.1 *Tetratrichomonas sp.*

10. Use the above data as an input into a "one click" phylogeny. Experiment with the other options such as "Advanced" or "A la Carte".

11. Based on the results of the phylogenetic analysis using the 18S rRNA sequences of the above 6 organisms, answer the following questions:

Of the six species, which is the most distantly related?

Of the 6 species, which species is the sister group to *Histomonas*?

Use the Activity 6 write up space below to keep any additional notes.

BIOINFORMATICS 4
ACTIVITY 7 – PART 1

In some cases, phylogenetic trees may be constructed from sequences other than the 18S rRNA sequences. Other gene sequences that have been used extensively are the haemoglobin genes (either of the alpha (α) or beta (β) globin genes).

Phylogeny based on the globin genes can only be used for organisms that have, or express, the haemoglobin genes.

This exercise is similar to the previous exercise, but, in this case, we will use the β globin sequences to construct the phylogenetic tree.

1. As before, using the NCBI database, locate the b globin gene sequence that has the accession number NM_000518.

Within the flat file obtained, determine whether the sequence present is an RNA sequence or a DNA sequence. How can you tell?

2. Display the sequence as a FASTA file by selecting the FASTA link at the top of the sequence.

3. In the FASTA file format, again note the header at the top of the sequence and copy the entire sequence entry, with header.

4. Using the BLASTN option, search the total nucleic acids database with the sequence entry from accession number NM_000518.

From the BLAST output file, name four other species with sequences similar to that of NM_000518.
1.
2.
3.
4.

Do the sequences from the four species share sequence similarity? If so, are they homologous? Why?

BIOINFORMATICS 4
ACTIVITY 7 – PART 2

For this part of the exercise, we will determine the genetic relationships between a number of different organisms based on their β-haemoglobin nucleotide sequences.

1. Find the following β-globin sequences in NCBI and cut and paste each one of them into a word file in a suitable format (i.e. in FASTA format).

   The β-globin sequences to look up are:

NM_000518	*Homo sapiens*
NM_173917	*Bos taurus*
J00860	*Gallus gallus*
V01433	*Xenopus laevis*
X69958	*Salmo salar*
NM_131020	*Danio Rerio*
X05080	*Rattus norvegicus*
NM_001278161	*Mus musculus*

10. Use the above data as an input into a "one click" phylogeny. Experiment with the other options such as "Advanced" or "A la Carte".

11. Based on the results of the phylogenetic analysis using the β-globin sequences of the above 8 organisms, answer the following questions:

Of the eight organisms, which are the most distantly related?

Of the 8 organisms, which are the most closely related?

ADDITIONAL NOTES SECTION

Include here any additional notes for this activity.

WEEK 8 - BIOINFORMATICS 5

REVISION AND ACTIVITIES UTILISING BIOINFORMATICS PRINCIPLES DEVELOPED WITHIN THE COURSE

BIOINORMATICS 5 ACTIVITY 1

Background

You work in a Forensics laboratory and have received some samples that have originated from a crime scene where a victim was found dead on the floor next to a floor mat. The victim had a wound to the head and bleeding had occurred. The following samples were collected from the crime scene for analysis, DNA extraction and sequencing.

Sample 1: a blood spot from the floor mat near the victim's body.

Sample 2: a saliva sample from a cigarette butt left in an ash tray at the crime scene.

Sample 3: fingerprint material from a can of soft drink on a table at the crime scene.

Sample 4: a grey coloured hair found on the floor mat near the victim's body.

Sample 5: an orange coloured hair found on the arm of a couch in the room where the victim's body had been found.

Sample 6: an dark coloured hair found on the seat of an armchair in the room where the victim's body had been found.

Sample 7: material from a dark bloody smear found on the glass of the window in the room where the victim had been found.

Sample 8: material from a bloody spot on the wall of the room in which the victim had been found.

You are able to extract DNA from all samples received and obtain DNA sequence data to help determine the origin(s) of the sample evidence. The sequencing data returned is as follows:

Sample 1 sequence, from the floor mat near the victim's body

```
cagcagatgt ggatcagcaa gcaggagtat gacgagtccg gcccctccat cgtccaccgc
aaatgcttct aggcggacta tgacttagtt gcgttacacc ctttcttgac aaaacctaac
ttgcgcagaa aacaagatga gattggcatg gctttatttg ttttttttgt tttgttttgg
ttttttttt ttttttggct tgactcagga tttaaaaact ggaacggtga aggtgacagc
agtcggttgg agcgagcatc ccccaaagtt cacaatgtgg ccgaggactt tgattgcaca
```

Sample 2 sequence, from saliva from the cigarette butt

```
cagcagatgt ggatcagcaa gccggagtat gacgagtccg gcccctccat cgtccaccgc
aaatgcttct aggcggacta tgacttagtt gcgttacacc ctttcttgac aaaacctaac
ttgcgcagaa aacaagatga gattggcatg gctttatttg ttttttttt tttgttttgg
ttttttttt ttttttttg gcttgactca ggatttaaaa actggaacgg tgaaggtgac
agcagtcggt tggagcgagc ccccccaaa gttcacaatg tggccgagga ctttgattgc
aca
```

Sample 3 sequence, from fingerprint material on a soft drink can at the crime scene

```
cagcagatgt ggatcagcaa gcaggagtat gacgagtccg gcccctccat cgtccaccgc
aaatgcttct aggcggacta tgacaaagtt gcgttacacc ctttcttgac aaaacctaac
ttgcgcagaa aacaagatga gattggcatg gctttatttt ttttttttgt tttgttttgg
ttttattttt ttttttggct tgactcagga tttaaaaact ggaacggtga aggtgacagc
agtcggttgg agcgagcatc cccccaagtt cacaatgtgg ccgggggctt tgattgcaca
```

Sample 4 sequence, from a grey coloured hair on the floor mat near the victim's body

```
gtccggcccc tccatcgtgc accgcaagtg cttctaggcg gactgttact gagctgcgtt
ttacaccctt tctttgacaa aacctaactt gcgcagaaaa aaaaaaaata agagacaaca
ttggcatggc tttgtttttt taaatttttt ttaaagtttt ttttttttt ttttttttt
tttttaagtt tttttgtttt gttttggcgc ttttgactca ggatttaaaa actggaacgg
tgaaggcgac agcagttggt tggagcaaac atcccccaaa gttctacaaa tgtggctgag
```

Sample 5 sequence, from an orange coloured hair found on the arm of a couch in the room

```
catcctgacc ctcaagtacc ccatcgaaca cggcattgtc accaactggg acgacatgga
gaagatctgg caccacacct tctacaacga gctgcgcgtg gcccccgagg agcacccggt
gctgctcacc gaggcccccc tgaaccccaa ggccaaccgt gagaagatga ctcagatcat
gttcgagacc ttcaacaccc cagccatgta cgtggccatc caggccgtgc tgtctctgta
cgcttctggc cgcaccactg gtattgtcat ggactctggg gacggggtca cccacactgt
```

Sample 6 sequence, from a dark coloured hair found on the seat of an armchair in the room

```
aagcggggca tcctgaccct caagtacccc atcgagcacg gcatcgtcac caactgggac
gacatggaga agatctggca ccacaccttc tacaacgagc tgcgcgtggc ccccgaggaa
gcacccgtgc tgctgaccga ggcccccctg aatcccaaag ccaaccgtga gaagatgact
cagatcatgt tcgagacgtt caacacccca gccatgtatg tggccatcca ggctgtgctg
tccctgtacg cctctggccg caccactggt attgtcatgg actctgggga tggcgtcacc
```

Sample 7 sequence, from the dark bloody smear found on the glass of the window in the room

```
cggccgtccc gtcatcaagg gtgtcatggg ttggtatggg acaaaaggac tcgtatgtcg
gtgatgaagc ccaaagcaaa cgtggtatcc tcacattgaa ataccccatt gaacacggta
ttgtcaccaa ctgggacgat atggaaaaga tctggcatca caccttctac aatgaattgc
gtgtcgcccc cgaagaacat cccgtcttgt tgactgaagc cccattgaat cccaaggcta
atcgtgagaa gatgacccaa atcatgttcg aaaccttcaa ctcccccgct atgtatgttg
```

Sample 8 sequence, from material from a bloody spot on the wall of the room

```
ggatctgttc gccaacaccg tcctgtccgg tggatctacc atgttcggtg gaatcgctga
ccgaatgcag cgcgaactga cggctctggc gccgccatcg atcaaggtca agatcgtggc
tccaccggag cgcaagtact cggtctggat cggtggctcc atcctcgcct cactgtccac
cttccagcag atgtggatct ccaagcagga gtacgacgag tccggccccg gaatcgtcca
ccgcaagtgt ttctaagctt acgtaatatc tgtattgttc gattgttaga ttgctcgttt
```

Question 1

Using BLAST searches, can you identify the origin of each of the sequences derived from each crime scene sample? Include your answers in the table below.

Crime scene sample	**Gene sequence**	**Organism**
Sample 1		
Sample 2		
Sample 3		
Sample 4		
Sample 5		
Sample 6		
Sample 7		
Sample 8		

Question 2

Which of the sample DNA sequences could have originated from either the victim or the perpetrator(s)?

Question 3

The following is the normal, or wild type, sequence (i.e. derived from the GenBank database) of the gene sequenced in all samples.

```
cagcagatgt ggatcagcaa gcaggagtat gacgagtccg gcccctccat cgtccaccgc
aaatgcttct aggcggacta tgacttagtt gcgttacacc ctttcttgac aaaacctaac
ttgcgcagaa aacaagatga gattggcatg gctttatttg ttttttttgt tttgttttgg
ttttttttct ttttttggct tgactcagga tttaaaaact ggaacggtga aggtgacagc
agtcggttgg agcgagcatc ccccaaagtt cacaatgtgg ccgaggactt tgattgcac
```

(A) Using BLAST with 2 sequences at a time, compare each of the potential victim and/or perpetrator sequences (from question 2) against the normal sequence provided.

(B) Looking at each of the BLAST outputs, are all the sequences identical to the normal sequence for the gene?

(C) How does each sequence differ from the normal or wild type sequence?

[i.e. do they have single base changes, sequence deletions or insertions, and how many changes are there for each sequence?]

Question 4

You have just been sent the following DNA sequence which is derived from a sample from the victim of the crime.

```
cagcagatgt ggatcagcaa gcaggagtat gacgagtccg gcccctccat cgtccaccgc
aaatgcttct aggcggacta tgacttagtt gcgttacacc ctttcttgac aaaacctaac
ttgcgcagaa aacaagatga gattggcatg gctttatttg tttttttttgt tttgttttgg
ttttttttt ttttttggct tgactcagga tttaaaaact ggaacggtga aggtgacagc
agtcggttgg agcgagcatc ccccaaagtt cacaatgtgg ccgaggactt tgattgcaca
```

Which of the 8 original sample sequences is the match for the victim's sequence? *[Hint: use BLAST searches.]*

Question 5

Based on your answer for Q4, which sample sequence(s) potentially correspond to the perpetrator(s)?

Question 6

You have just received information that the following sequence is derived from one of the policemen investigating the crime scene.

```
cagcagatgt ggatcagcaa gcaggagtat gacgagtccg gcccctccat cgtccaccgc
aaatgcttct aggcggacta tgacaaagtt gcgttacacc ctttcttgac aaaacctaac
ttgcgcagaa aacaagatga gattggcatg gctttatttt tttttttgt tttgttttgg
ttttatttt ttttttggct tgactcagga tttaaaaact ggaacggtga aggtgacagc
agtcggttgg agcgagcatc ccccaagtt cacaatgtgg ccgggggctt tgattgcaca
```

Does this sequence correspond to the sequences from any of the potential perpetrators? *[Hint: use BLAST searches.]*

Question 7

If yes, which of the original sample sequences does is correspond to?

Question 8

Which sample sequence most likely belongs to the perpetrator of the crime?

Question 9

From the original sample sequences, which samples could possibly represent samples from potential pets of the victim?

BIOINORMATICS 5
ACTIVITY 2

The problem

You have been given 2 primers (forward and reverse) which can be used for PCR amplification of a gene of interest. The sequences of the primers are shown below, however you have no information as to the gene or sequence region that is amplified by this primer pair, or the length of the PCR product that will be produced.

The primer sequences are as follows:

Forward primer sequence: `5′-attctgaggttatctttttaccacag-3′`

Reverse primer sequence: `5′-ggaaaaacatcaaaaaataacttacc-3′`

Based on the primer sequences, use your bioinformatics knowledge to determine the gene or sequence region amplified by these primers when used in a PCR reaction. What is the length of the PCR product produced?

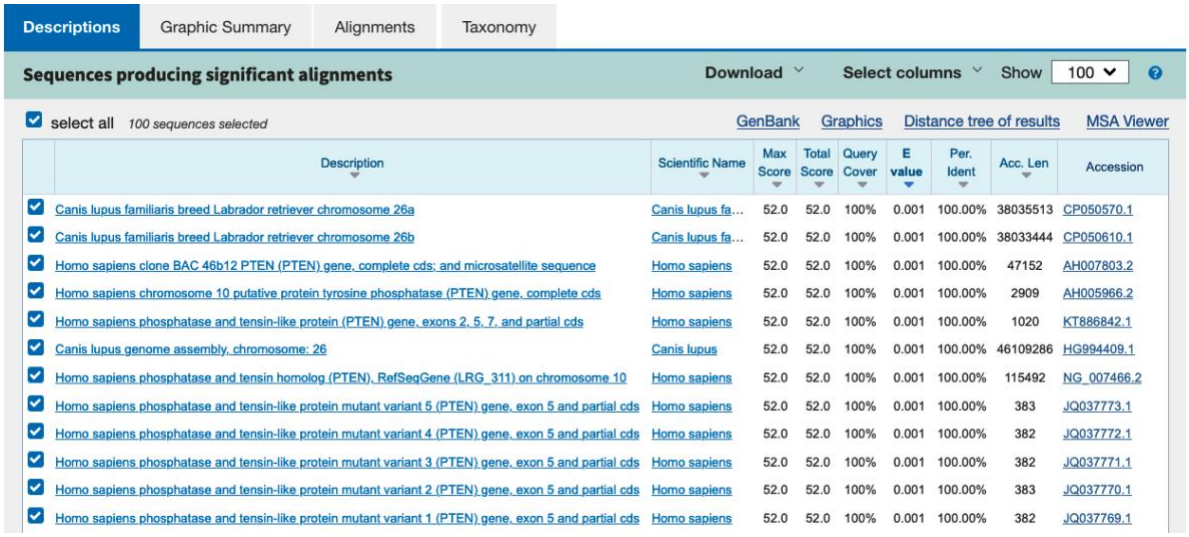
Approach to finding the answer

This exercise will be completed in a stepwise fashion.

Firstly, to determine what gene or region is amplified by these primers, conduct a BLAST search with each primer sequence.

Staring with the forward primer sequence, note down the top 5 sequence matches identified for this sequence.

As an example, the forward primer BLAST result is shown below and your search results should be the same.



Descriptions | Graphic Summary | Alignments | Taxonomy

Sequences producing significant alignments Download Select columns Show 100

select all *100 sequences selected* GenBank Graphics Distance tree of results MSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
Canis lupus familiaris breed Labrador retriever chromosome 26a	Canis lupus fa...	52.0	52.0	100%	0.001	100.00%	38035513	CP050570.1
Canis lupus familiaris breed Labrador retriever chromosome 26b	Canis lupus fa...	52.0	52.0	100%	0.001	100.00%	38033444	CP050610.1
Homo sapiens clone BAC 46b12 PTEN (PTEN) gene, complete cds; and microsatellite sequence	Homo sapiens	52.0	52.0	100%	0.001	100.00%	47152	AH007803.2
Homo sapiens chromosome 10 putative protein tyrosine phosphatase (PTEN) gene, complete cds	Homo sapiens	52.0	52.0	100%	0.001	100.00%	2909	AH005966.2
Homo sapiens phosphatase and tensin-like protein (PTEN) gene, exons 2, 5, 7, and partial cds	Homo sapiens	52.0	52.0	100%	0.001	100.00%	1020	KT886842.1
Canis lupus genome assembly, chromosome: 26	Canis lupus	52.0	52.0	100%	0.001	100.00%	46109286	HG994409.1
Homo sapiens phosphatase and tensin homolog (PTEN), RefSeqGene (LRG_311) on chromosome 10	Homo sapiens	52.0	52.0	100%	0.001	100.00%	115492	NG_007466.2
Homo sapiens phosphatase and tensin-like protein mutant variant 5 (PTEN) gene, exon 5 and partial cds	Homo sapiens	52.0	52.0	100%	0.001	100.00%	383	JQ037773.1
Homo sapiens phosphatase and tensin-like protein mutant variant 4 (PTEN) gene, exon 5 and partial cds	Homo sapiens	52.0	52.0	100%	0.001	100.00%	382	JQ037772.1
Homo sapiens phosphatase and tensin-like protein mutant variant 3 (PTEN) gene, exon 5 and partial cds	Homo sapiens	52.0	52.0	100%	0.001	100.00%	382	JQ037771.1
Homo sapiens phosphatase and tensin-like protein mutant variant 2 (PTEN) gene, exon 5 and partial cds	Homo sapiens	52.0	52.0	100%	0.001	100.00%	383	JQ037770.1
Homo sapiens phosphatase and tensin-like protein mutant variant 1 (PTEN) gene, exon 5 and partial cds	Homo sapiens	52.0	52.0	100%	0.001	100.00%	382	JQ037769.1

Step 1

What are the top 5 sequence matches for the forward primer? In table 1 below, include the gene or sequence region, the % identity and accession number of each of the top 5 sequence matches.

Table 1 Forward Primer sequence matches

Gene Sequence/Region	%Identity	Accession Number
1.		
2.		
3.		
4.		
5.		

Similarly, conduct a BLAST search with the reverse primer sequence and complete table 2 including the gene/sequence region, % identity and accession number for each of the top 5 matches.

Table 2 Reverse Primer sequence matches

Gene Sequence/Region	%Identity	Accession Number
1.		
2.		
3.		
4.		
5.		

Step 2

Of the top 5 sequence matches for each primer sequence, is there a sequence/region that shows sequence identity with both the forward and reverse primer sequences? If so, which sequence is it?

Select this sequence in the BLAST output for the forward primer search.

Looking at the match visually, note down the nucleotide numbers in the 'subject' sequence showing sequence identity with the 'query' sequence (i.e. the forward primer).

Query sequence region: 1 - 26

Subject sequence region:

Step 3

Now select the same sequence in the BLAST output for the reverse primer search.

Looking at the match, note down the nucleotide numbers in the 'subject' sequence showing sequence identity with the 'query' sequence (i.e. the reverse primer in this case).

Query sequence region: 1 – 26

Subject sequence region:

Step 4

Taking the subject sequence range for the forward primer and the subject sequence range for the reverse primer, can you work out the length of the PCR product? *[Hint: i.e. the number of nucleotides encompassed in the sequence ranges.]*

BIOINORMATICS 5
ACTIVITY 3

Determining genetic relationships between different organisms based on sequence of the beta actin gene.

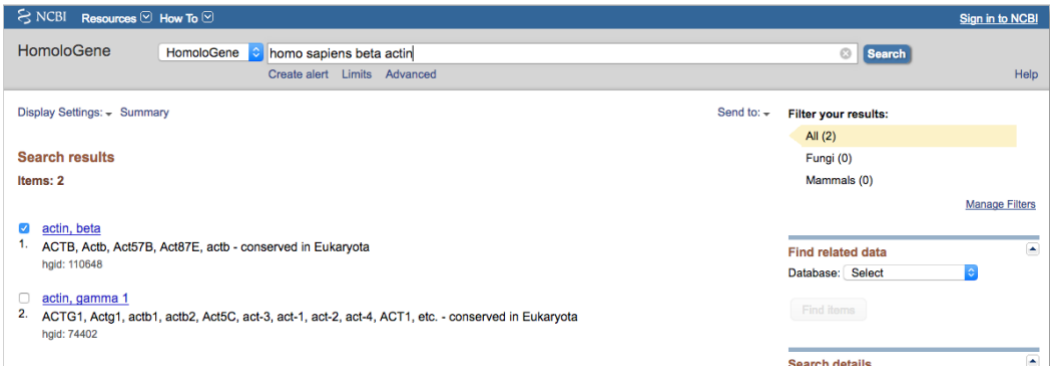
Beta actin is a cytoskeletal protein known to occur in many different organisms and is therefore a good protein sequence to use.

This exercise is similar the exercise you did last week but, instead of looking up each of the sequences individually, homoloGene will be used to find a set of protein sequences to compare for phylogenetic analysis.

2. Navigate to the NCBI search page and select homoloGene from the drop-down menu.

3. Now search for *homo sapiens* beta actin.

4. The results of the search will be seen as follows:



5. Select the first result by 'clicking' on it.

6. This will take you to the results page showing the details for both protein and nucleotide sequence data for beta actin from a wide variety of organisms.

   Selecting any of the nucleotide and protein accession numbers will take you to each of the individual flat files.

7. Select 'show multiple alignment' to view all the sequences aligned with each other showing the similarities and differences.

8. Selecting 'Download' from the top of the page will take you to a download screen where all sequences have been selected.

9. Pressing the 'Download' button will now download a text file with all sequences together (each in FASTA format) ready to be used for multiple sequence alignment (using ClustalO or Clustal Omega) and phylogenetic analysis.

10. Use the above data as an input into a "one click" phylogeny (see last week's notes).

    [Remember you can also generate a phylogenetic tree using Clustal Omega.]

11. Compare the results of the phylogenetic analysis using the beta actin sequences with that produced by using the β-globin sequences.

Are the most distantly related organisms the same in both phylogenetic analyses (or phylogenetic trees)?

Are the most closely related organisms the same in both phylogenetic analyses?

BIOINFORMATICS REVISION QUESTIONS

The following revision questions are to help you begin to revise your work in preparation for the upcoming bioinformatics practical test to be held in week 10.

Further revision questions will also be posted on Canvas at the conclusion of this week's classes.

Q1. Which of the following are literature databases?

a) MEDLINE and PubMED
b) MEDLINE and PDB
c) PubMED and PDB
d) MEDLINE and PDS

ANSWER: _____

Q2. Which of the following is a protein sequence database?

a) DDBJ
b) EMBL
c) GenBank
d) PIR

ANSWER: _____

Q3. Who maintains the GenBank nucleic acid sequence database?

a) Brookhaven laboratory
b) DNA database of Japan (DDBJ)
c) European Molecular Biology laboratory (EMBL)
d) National Centre for Biotechnology Information (NCBI)

ANSWER: _____

Q4. Which of the following is NOT a protein sequence databases?

a) PIR
b) PSD
c) SWISS PROT
d) EMBL

ANSWER: _____

Q5. What is the name of the information retrieval tool of NCBI GenBank?

a) Entrez
b) STAG
c) SeqIn
d) text search

ANSWER: _____

Q6. Which of the following is a sequence alignment tool?

a) BLAST
b) PRINT
c) PROSITE
d) PIR

ANSWER: _____

Q7. In what way can sequence alignment helps scientists?

a) to trace out evolutionary relationships
b) to infer the functions of newly synthesised genes
c) to predict new members of gene families
d) all of the above

ANSWER: _____

Q8. Which of the following is NOT a sequence alignment tool?

a) Rasmol
b) BLAST
c) FASTA
d) Clustal W

ANSWER: _____

Q9. Which of the following are applications of bioinformatics?

a) data storage and management
b) drug design
c) understanding relationships between organisms
d) all of the above

ANSWER: _____

Q10. Which of the following is a nucleotide sequence database?

a) EMBL
b) SWISS PROT
c) PROSITE
d) TREMBL

ANSWER: _____

Q11. You do a BLAST search on a DNA sequence and it identifies it as 'Exon 1' of a certain gene. What is an exon?

a) A section of a eukaryotic gene that is translated into protein.
b) A section of a eukaryotic gene that is NOT translated into protein.
c) A regulatory sequence that turns genes on and off.
d) DNA that has no genetic role but does maintain the physical structure of a chromosome.

ANSWER: _____

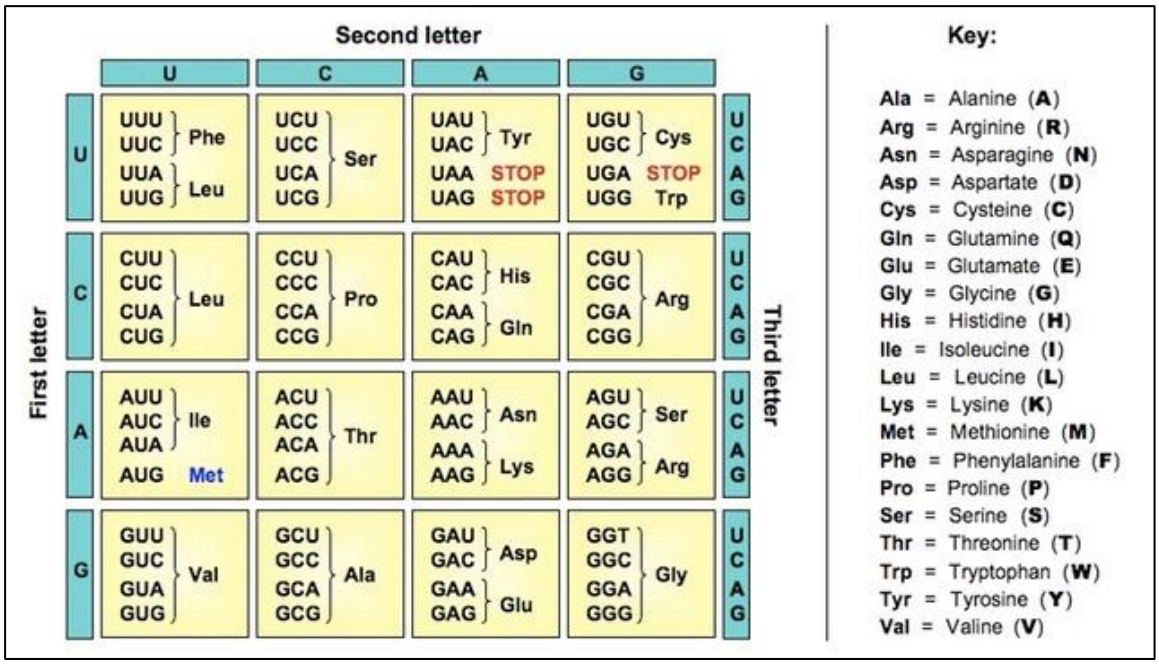
Q12. The procedure of aligning many sequences simultaneously is called…

a) multiple sequence alignment
b) pair wise alignment
c) global alignment
d) local alignment

ANSWER: _____

APPENDIX TO THE BIOINFORMATICS SECTION

APPENDIX 1: The Genetic Code and Amino Acid Abbreviations



APPENDIX 2: Applications of Bioinformatics in Various Areas of Biological Sciences

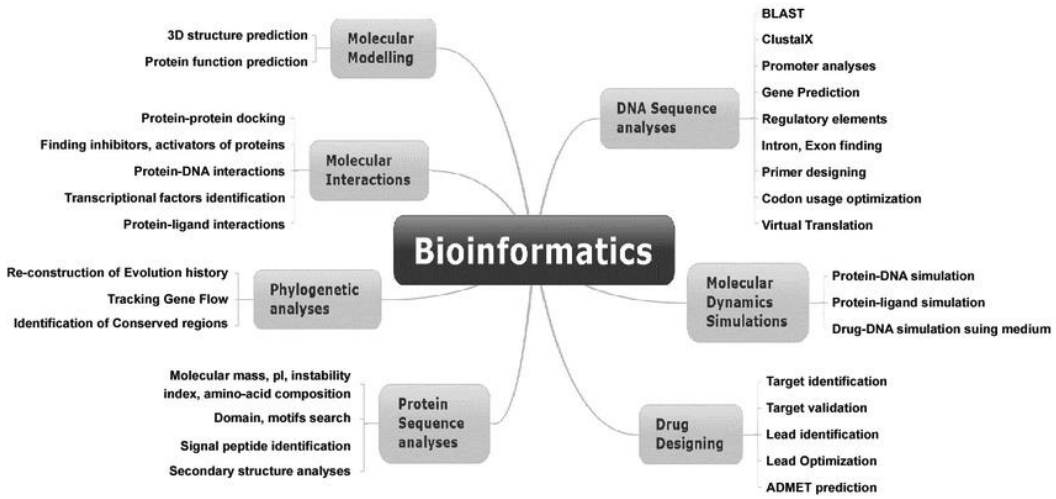


Figure 1: Application of bioinformatics tools in various areas of biological sciences.